
The Hump, Not the Cap: A Negative-Result Audit and a Structural Floor for the Stable Andrews–Curtis Conjecture on $AK(3)$

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 The Andrews–Curtis (AC) conjecture asserts that every balanced presentation of
2 the trivial group reduces to the trivial presentation by elementary moves; its stable
3 variant additionally permits stabilization by a generator. The Akbulut–Kirby
4 presentation $AK(3) = \langle x, y \mid x^3 = y^4, xyx = yxy \rangle$ has stood as the shortest
5 well-known potential counterexample since 1985. A 2024 claim that $AK(3)$ is
6 stably AC-trivial was rescinded in a later revision after a misprint was found in
7 the underlying construction, and a second, independently authored paper inherits
8 the same rescinded premise. We re-verify computationally that the stable status of
9 $AK(3)$ is open again (7 of 7 independent checks agree; two data defects found in a
10 published trivialization artifact), while confirming that the plain-AC equivalence
11 $AK(3) \sim P25$ survives via a certified 53-move replay. We then audit whether
12 stabilization opens shortcuts unreachable by 2-generator search. Three strategies
13 — a naive $z = w$ construction over 640 Miller–Schupp presentations, a 97-word
14 literature-grounded bank on $AK(3)$, and a five-lane campaign mining 180,645
15 distinct 2-generator quotients — record 0 of 16,870 solve attempts succeeding,
16 within the searched budgets (up to 10^6 nodes per attempt, beam width ≤ 2048).
17 We further show the $L = 24$ length cap is not the binding constraint in the tested
18 range ($L \leq 48$). Structurally, the length-13 floor of $AK(3)$ ’s AC-class has exactly
19 two canonical (signed-relabel) representatives, joined by a certified 21-move AC
20 path. Every finding ships as a machine-checkable certificate (20,947 checks, two
21 independent verifiers, 0 failures). A rigorously verified negative, plus a structural
22 map of where stabilized search stalls, gives the community a firm floor rather than
23 a claimed summit.

24 1 Introduction

25 The Andrews–Curtis (AC) conjecture Andrews and Curtis [1965] states that every *balanced* pre-
26 sentation of the trivial group — a presentation $\langle x_1, \dots, x_n \mid r_1, \dots, r_n \rangle$ with as many relators as
27 generators that presents the one-element group — can be transformed into the trivial presentation
28 $\langle x_1, \dots, x_n \mid x_1, \dots, x_n \rangle$ by a finite sequence of elementary moves. Sixty years after it was posed, it
29 remains open, and its difficulty is not merely combinatorial folklore: a balanced presentation that re-
30 sisted trivialization would refute the conjecture, and the same balanced presentations sit at the heart
31 of low-dimensional topology, where they encode handle decompositions of contractible 4-manifolds.
32 Potential AC counterexamples arise directly from Akbulut–Kirby’s smooth 4-dimensional construc-
33 tions Akbulut and Kirby [1985] and from candidate counterexamples to the Property 2R and slice-
34 ribbon conjectures Gompf et al. [2010], and the conjecture is explicitly entangled with the smooth
35 4-dimensional Poincaré problem Freedman et al. [2010]. Among all candidates, the Akbulut–Kirby

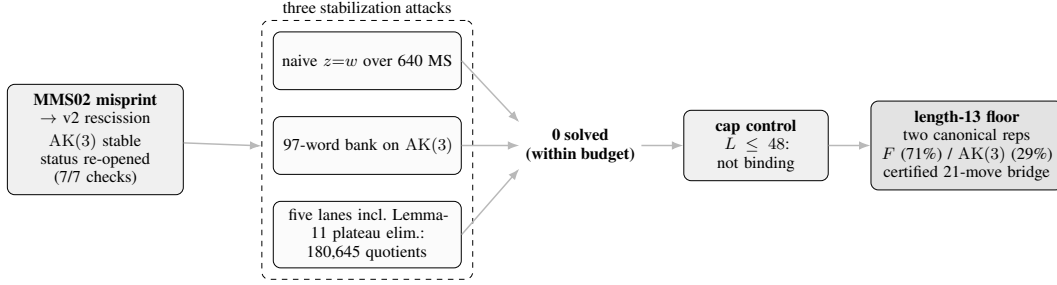


Figure 1: End-to-end audit pipeline: a greedy substitution solver, a stabilization move that defines a fresh generator z by a chosen word $w(x, y)$, a 97-word literature-grounded bank, and five attack lanes (meet-in-the-middle, direct stable-move search, trivial- z control, plateau elimination, and a zero-shot RL beam probe), with every reported solution emitted as a machine-checkable certificate replayed by two independent verifiers.

36 presentation $AK(3) = \langle x, y \mid x^3 = y^4, xyx = yxy \rangle$ is the shortest and most-studied potential
 37 counterexample, standing since 1985.

38 The *stable* (or *weak*) AC conjecture relaxes the problem by allowing two further moves, stabilization
 39 by a trivial generator–relator pair (AC4) and its inverse (AC5); a presentation reducible to the trivial
 40 one under AC1–AC5 is *stably AC-trivial*. The stable status of $AK(3)$ has recently oscillated. The
 41 first version of a 2024 case study Shehper et al. [2025] announced that $AK(3)$ is stably AC-trivial,
 42 removing it as a stable-AC counterexample; a subsequent revision located a misprint in the Wirtinger
 43 construction of Myasnikov et al. [2002] on which that argument depended and rescinded the claim.
 44 A parallel automated-deduction paper Lisitsa [2025], written between those two versions, builds its
 45 stable-triviality conclusion on the same, now-rescinded premise and has not been revised. The plain
 46 reading of the current literature is therefore blunt: the stable AC-triviality of $AK(3)$ is open again.

47 Search-based and learning-based methods have crossed hard barriers in this domain before — ge-
 48 netic algorithms trivialized $AK(2)$ Miasnikov [1999], and a recent study of the AC difficulty land-
 49 scape Fagan et al. [2026] pushed reinforcement learning past presentations that classical search
 50 cannot reach. This invites a concrete question about stabilization as an escape hatch: does adding
 51 a generator z defined by a relator $z^{-1}w(x, y)$ — for a cleverly chosen word w in the original gen-
 52 erators — open shortcuts (informally, “wormholes”) through the enlarged AC graph that ordinary
 53 2-generator search cannot find? If some w trivialized $AK(3)$ where the dumb choices $z = r_i$ do not,
 54 stabilization would be a genuine algorithmic lever rather than bookkeeping.

55 This paper is a systematic audit of that hope, and it returns a rigorously verified negative paired
 56 with a structural map (Figure 1). We first re-verify, seven independent ways, that the stable status of
 57 $AK(3)$ is open while its plain-AC equivalence to the length-25 presentation P25 survives. We then
 58 run three stabilization strategies of increasing sophistication — all of which fail within the searched
 59 budgets — and a control showing that the per-relator length cap is not what stops them. Finally, we
 60 characterize *where* stabilized greedy search stalls: a length-13 floor whose AC-class has exactly two
 61 canonical representatives, bridged by a certified move sequence. Throughout, every positive claim
 62 is emitted as a machine-checkable certificate.

63 Contributions.

- 64 • **Re-verification of a re-opened question.** We computationally confirm, via 7 of 7 agree-
 65 ing independent checks (Smith normal form, Todd–Coxeter coset enumeration, and S_3 -
 66 homomorphism counting among them), that $AK(3)$ ’s stable status is open, that the plain-
 67 AC link $AK(3) \sim P25$ survives (53-move replay, plus validation of an independent Prover9
 68 derivation), and we surface two data defects in a published trivialization artifact.
- 69 • **Systematic stabilization negatives at stated budgets.** Across a naive $z = w$ construction
 70 over 640 Miller–Schupp presentations, a 97-word literature-grounded bank on $AK(3)$, and
 71 a five-lane campaign that mines 180,645 distinct 2-generator quotients, 0 of 16,870 solve
 72 attempts succeed within the searched budgets (up to 10^6 nodes per attempt, beam width
 73 ≤ 2048).

- 74 • **A cap-not-binding control.** We show the $L = 24$ per-relator length cap is not the binding
75 constraint in the tested range: relaxing to $L \leq 48$ (and total length ≤ 60) does not convert
76 a single failure into a success.
 - 77 • **A two-representative structural floor.** The length-13 floor of $\text{AK}(3)$'s AC-
78 class has exactly two canonical (signed-relabel) representatives — $F = \langle x, y \mid$
79 $y^{-2}xyx^{-2}, y^{-3}x^{-2}yx \rangle$, a 2-generator presentation AC-equivalent to $\text{AK}(3)$ (712/1006
80 = 71% of floor states), and $\text{AK}(3)$'s own reduced form (294/1006 = 29%) — joined by a
81 certified 21-move AC path.
 - 82 • **A certificate framework and reusable archive.** We release the `acx-cert-v1` machine-
83 checkable certificate format with two independently authored verifiers (20,947 checks, 0
84 failures) and the full candidate-quotient archive, so every claim above is independently
85 re-runnable.
- 86 Code, data, and all certificates will be released on acceptance.

87 2 Background and Related Work

88 2.1 AC moves, substitution, and stable AC

89 A *balanced presentation* $\langle x_1, \dots, x_n \mid r_1, \dots, r_n \rangle$ has equally many generators and relators, each
90 relator a word in the $x_i^{\pm 1}$; throughout we write words with $x^{-1}yx$ notation and use the shorthand
91 $X = x^{-1}$, $Y = y^{-1}$ where it aids readability. The elementary Andrews–Curtis moves act on the
92 ordered tuple of relators: (AC1) replace some r_i by r_i^{-1} ; (AC2) replace some r_i by $r_i r_j$ with $j \neq i$;
93 (AC3) replace some r_i by $w r_i w^{-1}$ for an arbitrary word w . Two presentations related by a finite
94 sequence of AC1–AC3 are *AC-equivalent*, and a presentation AC-equivalent to the trivial one is
95 *AC-trivial*. Because the elementary moves take small steps, effective search operates on *substitution*
96 *supermoves* that package AC2 and AC3 into a single action: viewing both relators as cyclic strings,
97 a substitution rotates each and splices one into the other so that at least one pair of letters cancels
98 Fagan et al. [2026]. Substitutions enlarge the action space but shrink the effective state space, and
99 they are the move set our search inherits.

100 The *stable* (or *weak*) AC conjecture adds two moves Shehper et al. [2025]: (AC4) introduce a
101 fresh generator together with the trivial relator equal to it, replacing $\langle x_1, \dots, x_n \mid r_1, \dots, r_n \rangle$ by
102 $\langle x_1, \dots, x_n, x_{n+1} \mid r_1, \dots, r_n, x_{n+1} \rangle$; and (AC5), its inverse, deleting a generator and its trivial
103 relator. Presentations related by AC1–AC5 are *stably AC-equivalent*, and the stable conjecture as-
104 serts every balanced presentation is stably AC-trivial. Stabilization is exactly the freedom our audit
105 probes, and its most useful consequence here is Lemma 11 of Shehper et al. [2025] (substitution-
106 and-removal): given a trivial-group presentation $\langle x_1, \dots, x_n, y \mid r_1, \dots, r_n, y^{-1}w \rangle$ in which w is a
107 word in the x_i *only*, the n -generator presentation $\langle x_1, \dots, x_n \mid r'_1, \dots, r'_n \rangle$ obtained by substituting
108 w for every occurrence of y is stably AC-equivalent to it. The proof exhibits w as a product of
109 conjugates of the r'_i , but places no bound on how many such factors are needed, so the packaged
110 supermove has no *a priori* length bound (full statement in Appendix A).

111 2.2 The $\text{AK}(n)$ and Miller–Schupp families

112 The Akbulut–Kirby series $\text{AK}(n) = \langle x, y \mid x^n = y^{n+1}, xyx = yxy \rangle$ arose from smooth 4-
113 manifold topology Akbulut and Kirby [1985]; $\text{AK}(2)$ is AC-trivial, while every $\text{AK}(n)$ with $n > 2$
114 is an open case, $\text{AK}(3)$ being the shortest. The Miller–Schupp series $\text{MS}(n, w) = \langle x, y \mid x^{-1}y^n x =$
115 $y^{n+1}, x = w \rangle$, parameterized by $n \geq 1$ and a word w of exponent sum 0 in x , is a second
116 two-parameter family of balanced trivial-group presentations Miller and Schupp [1999], and it sup-
117 plies the 1190-presentation benchmark our sweeps use. These families are linked: $\text{AK}(n)$ is AC-
118 equivalent to $\text{MS}(n, w_1)$ with $w_1 = y^{-1}x^{-1}yxy$ (Proposition 5 of Shehper et al. [2025]). Two fur-
119 ther results ground our word bank. The isolation family $w_k = y^{-k}x^{-1}yxy$ threads a one-parameter
120 path of AC-equivalent presentations through $\text{MS}(n, w_k)$ for all $k \in \mathbb{Z}$ (Theorems 6 and 7 of Shehper
121 et al. [2025]), and the distinguished word $w_\star = y^{-1}xyx^{-1}$ makes $\text{MS}(n, w_\star)$ AC-trivial for every
122 n (Theorem 3). Each identifies structurally motivated words w worth throwing at stabilized $\text{AK}(3)$.

2.3 The misprint that re-opened the question

The stable-triviality argument for $AK(3)$ traces back to Theorem 1.4 of Myasnikov et al. [2002] (MMS02), which builds a 3-generator family of trivial-group presentations by deleting one redundant relator from a Wirtinger presentation of an unknot diagram and adjoining a trivializing relator. As printed, the 13th Wirtinger relator reads $x_{13} = x_5x_{12}x_5^{-1}$, whereas the diagram requires $x_{13} = x_4x_{12}x_4^{-1}$. The distinction is not cosmetic: in a genuine Wirtinger presentation every single relator is redundant, so deleting any one leaves the same knot group ($\mathbb{Z}$ for the unknot), but in the misprinted presentation W' different single-relator deletions give *non-isomorphic* groups — one deletion fingerprints as the three-strand braid group B_3 , another as $\mathbb{Z}$ Shehper et al. [2025]. Consequently the derived 3-generator family need not present the trivial group, and the stable AC-triviality of the length-25 presentation P25 obtained from it — and hence of $AK(3)$ — is unsupported. What survives is the *plain*-AC equivalence: P25 is AC-equivalent to $AK(3)$ regardless of the misprint. A 53-move path witnessing this is given in Appendix F of Shehper et al. [2025], and an independent Prover9 derivation of the same equivalence is archived in Lisitsa [2025, 2024b]; our audit re-verifies both (Section 4, Appendix E).

2.4 Related computational work

Computational attacks on AC date to exhaustive breadth-first enumeration, which classified all presentations up to length 13 Havas and Ramsay [2003], and to heuristic and metaheuristic searches Bowman and McCaul [2006], Miasnikov [1999], the latter trivializing $AK(2)$ by genetic search. Automated deduction has more recently produced new trivializations across the Miller–Schupp series Lisitsa [2024a], and machine learning has entered the arena from several directions: reinforcement learning over the AC graph Fagan et al. [2026], Lean-verified LLM-assisted search Zhang et al. [2025], and the closely related unknotting problem Gukov et al. [2021]. A crucial caveat frames every negative below. Balanced presentations of the trivial group can require trivialization sequences of superexponential length Bridson [2015], Lishak [2017], so no bounded-budget search can distinguish “no trivialization exists” from “the trivialization is longer than we looked”; our unsolved counts are therefore always reported within explicit budgets. Where prior work searches for trivializations, this paper audits a rescinded stable-triviality claim, quantifies the failure of stabilization to help within stated budgets, and characterizes the structural floor at which stabilized greedy search stalls — with every claim carried by a machine-checkable certificate.

3 Methods

Our campaign attacks a single target — showing $AK(3)$ to be stably Andrews–Curtis (AC) trivial — with a family of search methods that share one core solver and one certificate format. Figure 1 sketches the pipeline: a greedy substitution engine (§4 reports its behavior), a stabilization move that adds a generator defined by a chosen word, a literature-grounded bank of such words, and five attack lanes that layer stable-AC moves on top of the base search. Every reported solution is a machine-checkable certificate replayed by two independent verifiers. Terminology used below is collected in A.

3.1 Greedy substitution search and cap semantics

Our base solver reimplements the GS-Sub baseline of Fagan et al. [2026]: a best-first search over balanced presentations whose priority is the total relator length. The expansion is the *substitution* composite move. For every unordered pair of relators $\{a, b\}$ in a presentation, each candidate $c \in \{r_b, r_b^{-1}\}$, and every pair of cyclic rotations of r_a and c whose boundary letters cancel, the move emits the freely and cyclically reduced concatenation into both slots a and b . Relators are stored as signed integers under the encoding $x \rightarrow 1, X \rightarrow -1, y \rightarrow 2, Y \rightarrow -2, z \rightarrow 3$, with 0 as padding. States are keyed by a canonical form: the rotation- and inversion-invariant minimal representative of each relator under the letter order $Z < z < Y < y < X < x$ (inverse before generator, higher generator id first), so that AC-equivalent presentations collapse to one node.

We fix vocabulary precisely. A presentation with n generators is **solved** when the search reaches a presentation of total relator length n — the trivial presentation for n generators — *and* the recorded move path replays cleanly through an independent verification gate; the trivial target thus depends

174 on generator count, being total length 2 for a 2-generator state and 3 for a 3-generator state. When
 175 a run does not solve, we report its **floor**: the minimum total relator length reached within the node
 176 budget. Node budgets range from 10^5 to 10^6 per attempt.

177 Cap semantics are load-bearing and differ from the original. The GS-Sub baseline caps the *sum*
 178 of relator lengths at 100; our reproduction and every stabilized arm instead cap *each* relator at
 179 $L = 24$ (a per-relator cap), which removes a 2- versus 3-generator confound that a shared sum cap
 180 would introduce once a third relator is present. Under the per-relator cap, 634 of the 640 known-
 181 solvable MS(1190) presentations Miller and Schupp [1999] solve at 10^6 nodes; the remaining 6
 182 (indices 634–639) solve only under the original sum cap. This is a cap-semantics difference, not a
 183 reproduction failure. The per-node expansion hot path is numba-accelerated and held byte-identical
 184 to the reference Python implementation through differential gates on real visited sets (D).

185 3.2 Stabilization with a defined generator

186 The stabilization move introduces a generator z *defined* by a word $w(x, y)$:

$$\langle x, y \mid r_1, r_2 \rangle \longrightarrow \langle x, y, z \mid r_1, r_2, z \cdot w(x, y)^{-1} \rangle,$$

187 an AC4 stabilization followed by realizing the definition $z = w$; the added relator is the cyclically
 188 reduced word $[z] ++ \text{inverse}(w)$. This is a legal stable-AC move for *any* word w . The wormhole
 189 hypothesis motivating it is that a well-chosen w shares x, y -letters with r_1, r_2 and so may open
 190 substitution paths the plain 2-generator search cannot reach; destabilizing back out via a Lemma-11
 191 elimination would even certify plain AC-triviality of the original presentation. Two guards keep
 192 the search honest. A *null-revert block* pre-seeds the state in which the z -relator collapses to the
 193 bare generator z , blocking trivial reversal of the stabilization while keeping genuine destabilization
 194 reachable; blocked hits are counted rather than silently dropped. Every solved path is replayed by
 195 an independent path verifier before it is counted, and every 3-generator solve’s full move-and-state
 196 path is persisted as a first-class artifact for re-audit.

197 3.3 Literature-grounded word bank

198 For each AK(3) form we build a bank of 97 candidate words w — seven families plus two con-
 199 trols (Table 1). The families, in escalation priority from the strongest theory prior, are: **relhalf** (17
 200 words: sides, rotations, and inverses of the target’s own relators, e.g. xyx, yxy, x^3, y^4); **wk** (17:
 201 $y^{-k}x^{-1}yxy$ for $k \in [-8, 8]$, the isolation words of Theorems 6 and 7 of Shehper et al. Shehper et al.
 202 [2025]); **wstar** (5: $y^{-1}xyx^{-1}$ and its automorphism images, Theorem 3); **conj** (14: short conjugates
 203 $g x g^{-1}, g y g^{-1}$); **comm** (6: commutators and short double commutators); **ms** (3: MS(n, w) library
 204 words); and **brute** (33: all freely reduced words of length ≤ 3). The two **control** words are the
 205 target’s own relators r_1, r_2 , added at run time. Words are deduplicated by the *induced z -relator* and
 206 dropped unless that relator satisfies the per-relator cap $L = 24$. We sweep two AK(3) forms: the
 207 textbook presentation $\langle x, y \mid xyx = yxy, x^3 = y^4 \rangle$ and the dataset representative form ($Y X y X Y x$
 208 / $Y Y Y X X X X$). A two-tier protocol screens every word at 10^5 nodes in parallel, then escalates
 209 each unsolved word to 10^6 nodes in a sequential, memory-bound pass.

210 3.4 Hard-but-solvable controls

211 AK(3) returns an uninformative 0-everywhere signal, so a null result there cannot distinguish “no
 212 word helps” from “no method could help.” We therefore run the identical word-bank protocol on two
 213 MS(1190) presentations that the 2-generator baseline *does* solve, but only after substantial search
 214 — index 625 (77,385 nodes, 663 moves) and index 610 (61,066 nodes, 307 moves). These give a
 215 real, beatable reference: the dumb control $z = r_1$. Here relhalf is re-derived from each target’s own
 216 relators. The question these controls pose is sharp: does any literature-grounded word beat, or even
 217 match, the dumb control?

218 3.5 Five attack lanes for stable trivialization

219 **Lane A (meet-in-the-middle).** A catalog of 334 presentations obtained from a *corrected* Wirtinger
 220 unknot presentation via randomized Lemma-11 cascades, each stably AC-trivial by construction and
 221 each carrying a certificate; we search from AK(3) toward the catalog and in reverse. A pre-check

Table 1: The 97-word literature-grounded bank thrown at stabilized AK(3): seven families (95 words) grounded in AK(n)/Miller–Schupp theory or brute enumeration, plus two per-target control words $z = r_1, r_2$ added at run time.

Family	Count	Description	Theory grounding
relhalf	17	Relator halves/rotations/inverses of the target’s relators	Dumb baseline (no theorem)
wk	17	$y^{-k} x^{-1} y x y$ isolation words	Thm 6/7 (arXiv:2408.15332)
wstar	5	$y^{-1} x y x^{-1}$, plus automorphism images	Thm 3 (arXiv:2408.15332)
conj	14	Short conjugates $g x g^{-1}, g y g^{-1}$	Dumb baseline (no theorem)
comm	6	Commutators and doubles	Dumb baseline (no theorem)
ms	3	MS(n, w) library words	Miller-Schupp family
brute	33	All freely-reduced words of length ≤ 3	Exhaustive enumeration (no theorem)
control	2	The target’s own relators r_1, r_2	Sanity control ($z=r_i$)
Total	97		

Counts for the 7 literature-grounded families (95 words total) come from word_bank.json’s by_family; the 2 control words ($z=r_1, z=r_2$) are added at runtime and counted here from runs/ak3_rep_100000.jsonl’s family={'control'} rows.

Verification: family counts sum to 97, matching the 97-row ak3_rep_100000.jsonl (one row per word per form) (match).

classified all 151 in-cap catalog leaves as plain-AC-trivial by 2-generator greedy, which deprioritized this lane — a meeting point here would have implied the stronger claim of plain AC-triviality of AK(3).

Lane B (direct stable-move search). A best-first search whose move set is the *union* of substitution, stabilization $z = w$ over the word bank, and Lemma-11 elimination, over presentations of variable generator count up to a maximum, with priority equal to total length plus a penalty times the number of extra generators. As validation, known-easy instances solve *through* the stabilize/eliminate moves — for example a corrected-family instance in 3 nodes, with certificate.

Lane C (trivial- z control). Stabilization with the *empty* word (a bare generator z) at $n = 3, 4, 5$: pure stabilization without content, the dumb baseline against which content-bearing words are judged.

Lane D (plateau elimination — the methodological novelty). Stabilized greedy runs plateau because greedy has no destabilize move, yet every *visited* 3-generator state in which some generator occurs exactly once in some relator admits a Lemma-11 elimination to a 2-generator quotient that is stably AC-equivalent to AK(3) by construction of the harvest pipeline (verified end-to-end on the exported certificates). The pipeline harvests the full visited set of each run (up to $\sim 10^7$ states), eliminates every once-occurring generator, deduplicates by signed-relabel together with rotation/inversion symmetry, solves the resulting fresh 2-generator candidates shortest-first with plain greedy, and on any solve reconstructs and verifies the full certificate chain (stabilize $\rightarrow$ substitution path $\rightarrow$ eliminate $\rightarrow$ trivialization). At scale, the campaign harvested 180,645 distinct quotients across forms, words, and boxes, and ran 16,870 solve attempts over 6,058 distinct candidates.

Lane E (zero-shot RL transfer probe). A pretrained policy plus beam-search system, trained on 2-generator MS presentations Fagan et al. [2026], is run zero-shot on the stabilized and floor states (beam width 512–2048, with temperature annealing). This is explicitly an out-of-distribution probe of a fixed policy, not an in-principle negative result about reinforcement learning for this task.

Alongside the five lanes we run a **cap-control experiment**: search, harvest, and solve are re-run with L raised from 24 to 48 (and long stratified candidates solved at maximum length 60) to test directly whether the per-relator cap ever binds.

3.6 Certificates and independent verification

Every solution and every stable-AC equivalence is emitted as a JSON certificate (schema in B): an ordered list of steps, the full intermediate states, and the endpoint claims. Verification checks each

step’s replay and preconditions and enforces global invariants across every state — balancedness, and $|\det| = 1$ of the abelianization matrix, computed by exact integer Bareiss elimination. Two verifiers run on every certificate: the engine author’s, and an *independently authored* black-box verifier written from the specification alone, with its own reduction, canonicalization, determinant, and replay logic. Across the campaign these performed 20,947 checks with 0 failures; all real certificates pass both verifiers, and mutation-sensitivity is validated (corrupted steps, states, and targets are rejected). A separate literature-verification suite passes 7/7 independent checks — Smith-normal-form abelianization, Todd–Coxeter coset enumeration, and S_3 -homomorphism counting with a CSP-versus-brute-force self-test — reproducing the consequences of the presentation misprint that leaves stable triviality open. The printed 53-move Appendix-F path replays exactly from P25 to AK(3), and Lisitsa’s published Zenodo sequence S2 validates at cyclic-word granularity after bridging two data defects (one corrupt transition, one mislabeled conjugation direction) Lisitsa [2024b].

3.7 Compute disclosure

Development and calibrated runs were done on a consumer laptop (16 GB RAM); the five-lane escalation ran on cloud CPU boxes (up to ~ 50 GB RAM, 8 vCPU). Per-attempt node budgets ranged from 2.5×10^4 to 10^6 , and the campaign totals roughly 10^7 search nodes across all attempts, wall-clocked over about one week. No GPUs were used except for the pretrained-policy beam runs, which we executed as CPU-only inference in this campaign. Memory is the binding constraint — a greedy run’s visited set holds roughly $40\times$ its node budget in entries — and numba acceleration gives a $\sim 3.7\text{--}5.7\times$ speedup on the per-node hot paths.

4 Experiments and Results

Every search below shares the greedy substitution solver, the stabilization move, and the certificate format described in the methods; we report negatives strictly within the searched budgets and emit every positive as a machine-checkable certificate (Appendix B). We first re-verify the literature status of AK(3), then run stabilization strategies of increasing sophistication — all failing within the searched budgets — add a control that the length cap is not what stops them, and finally give a structural map of where the search stalls. Two scales recur and must not be conflated: a fully local reproduction (a 1,006-candidate floor census, 1,937 solve attempts) and the consolidated campaign archive (180,645 harvested quotients, 16,870 attempts, 6,058 distinct candidates); we keep them separate throughout.

4.1 Re-verifying the literature status of AK(3)

Table 2 collects seven independent computational checks of the misprint story, all 7/7 passing (Smith-normal-form abelianization, Todd–Coxeter/Felsch coset enumeration, and S_3 -homomorphism counting whose CSP hom-counter is validated 4/4 against brute force on synthetic triple-relator systems). The *corrected* Wirtinger presentation W is genuine: deleting any single relator — all 14 deletions — leaves a presentation of $\mathbb{Z}$. The misprinted W' provably is not: one single-relator deletion yields a B_3 fingerprint (12 homomorphisms to S_3 , 6 of them non-abelian), while a different deletion is $\mathbb{Z}$ -consistent — two deletions presenting *non-isomorphic* groups, which no Wirtinger presentation of any diagram can do. AK(3) itself presents the trivial group (coset enumeration, order 1), and the Wirtinger-derived M_3 presentation reduces under Lemma 11 with $z := y^{-1}x$ exactly to P25, pinning the commutator convention (convention B) of MMS02 Myasnikov et al. [2002]. The printed 53-move Appendix-F sequence Shehper et al. [2025] replays exactly from P25 to AK(3) (53 steps, 54 states), reaching a peak total relator length of 25 en route — above the length-13 floor of §4.8. An independently published trivialization Lisitsa [2024b, 2025] (159 transitions) validates at cyclic-word granularity once two data defects are handled (one corrupt transition, bridged by composing adjacent moves; one mislabeled conjugation direction). The consequence is unambiguous: the plain-AC bridge AK(3) $\sim$ P25 is solid, while the claimed stable-triviality of AK(3) has no surviving support (Appendix E).

Table 2: Seven independent computational checks (rows 1–7) of the MMS02 misprint account, all passing, plus two certificate/log re-verifications (rows 8–9): the corrected Wirtinger presentation is genuine, the misprinted W' is not, and the plain-AC bridge $P25 \sim AK(3)$ survives.

Check	Method	Result
Self-test: counter	hom- CSP hom-counter vs. brute-force cross-check on 4 synthetic triple-relator systems ($n \leq 4$ generators)	PASS (4/4 match)
Check 1: corrected-W, all 14 deletions	1: SNF abelianization + Felsch coset enumeration on each single-relator deletion of corrected W	PASS (14/14 present Z)
Check 2: misprinted-W is broken	2: S3 hom-counting gate: {7,14} disagree (B3 vs Z) while corrected-W deletions {7,12,14} all agree (Z)	PASS (gate satisfied)
Check 3: trivial	3: AK(3) is Felsch coset enumeration of the trivial subgroup (in-dex = group order)	PASS (order 1)
Check 4: (Wirtinger) = P25	4: M3 Reduced-word match of the Wirtinger-derived M3 transcription against P25 under commutator convention B	PASS (convention B matches)
Check 5: corrected-W, 3-gen reduction	5: SNF + S3 hom-count + coset collapse of corrected-W's (x,y,z) 3-generator reduction	PASS (presents Z)
Check 6: Remark 17 conjugation convention	Remark Conjugation-convention match ($g \ t \ g^{-1}$) of r1/r2 against Remark 17	PASS (r1, r2 both match)
Appendix-F move replay [cert]	53- Replay of the printed AC' h-move sequence (Appendix F, arXiv:2408.15332v2), forward or-der/forward moves, from P25	Lands exactly on AK(3) (53 steps, 54 states; cert appendixF_P25_to_AK3.json)
Lisitsa S2 external replay [log]	Independent replay of Zenodo 14567743's 159-transition AC path $P25 \rightarrow AK(3)$	PASS — genuine bridge; 2 data defects found in the published artifact (line 91 corrupt relator, line 81 CONJ label swapped)

Rows 1-7 are read verbatim from results/stable_ac/ak3_stable_proof/literature_checks.json (7 entries, all pass:true). Raw JSON key names differ in spelling/case from this table's human-readable labels; see the run report for the exact key list.

Rows 8-9 are additional re-verifications NOT drawn from literature_checks.json: row 8 is sourced from the certificate results/stable_ac/ak3_stable_proof/certs/appendixF_P25_to_AK3.json (steps count read live from the cert); row 9 is sourced from the campaign log experiments/stable_ac/ak3_stable_proof/RESULTS.md, section 'PL(partial) – Lisitsa S2 independently validated'.

Key-name check: this table's originally-expected key spellings ['check3_ak3_trivial', 'check4_M3_lemma11_P25', 'check5_corrected_family_Z'] do not appear verbatim in literature_checks.json; the actual keys (used above) are ['check3_AK3_trivial', 'check4_M3_to_P25', 'check5_correctedW_3gen'].

301 4.2 Naive $z = w$ stabilization does not extend the baseline

302 Our reference point is the 640 known-solvable MS(1190) presentations Miller and Schupp [1999]
303 (634 solve under the per-relator cap $L = 24$ at 10^6 nodes; the remaining 6 only under the
304 original sum cap). We stabilize each with a single defined generator $z = w$ and re-solve, for
305 $w \in \{r_1, r_2, x, y\}$, at 5×10^5 nodes (Figure 2, Table 3). Solved counts fall monotonically — 619
306 (r_1), 602 (r_2), 540 (x), 523 (y) — with 21/38/100/117 attempts budget-exhausted. Crucially, every
307 arm's solved set is a *strict subset* of the baseline's: computed row-by-row, $|\text{arm} \setminus \text{baseline}| = 0$ for all
308 four, the union across arms is 620 (below the baseline 634) and the intersection is 523. Stabilization
309 with a defined generator thus reproduces, but never extends, the baseline within the searched bud-
310 gets. Median node counts stay small (baseline 11; r_1, r_2 each 13; x, y each 10), yet the x and y arms'
311 *mean* nodes inflate to 14,681 and 21,833: defining z by an existing generator is a length-neutral gen-
312 erator swap that spawns alias orbits the search must exhaust before it terminates. No wormhole
313 appears where a defined z rescues a presentation the plain 2-generator search cannot already solve.

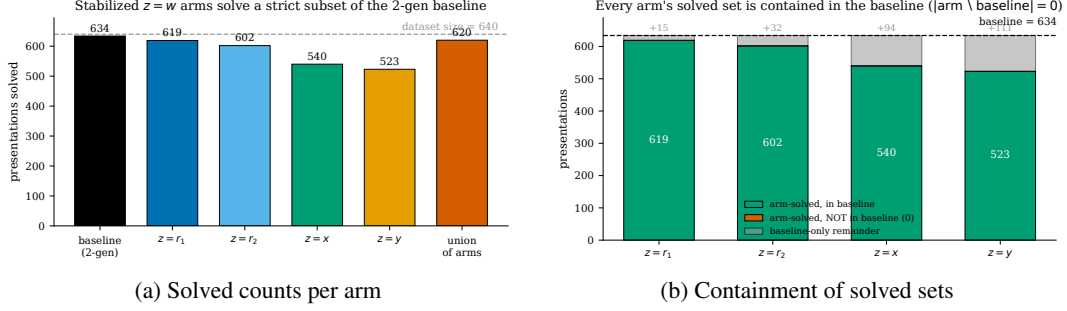


Figure 2: Naive $z = w$ stabilization over the 640 known-solvable MS(1190) presentations at 5×10^5 nodes. (a) Solved counts per arm — baseline 634, r_1 619, r_2 602, x 540, y 523 — all below the baseline. (b) Containment: every arm’s solved set is a strict subset of the baseline’s ($|\text{arm} \setminus \text{baseline}| = 0$ for all four arms; union 620, intersection 523), so stabilization reproduces but never extends the baseline within budget.

Table 3: Naive $z = w$ stabilization over the 640 known-solvable MS(1190) presentations at 5×10^5 nodes. Every arm’s solved set is a strict subset of the baseline’s; node and path-length statistics are computed over solved presentations only.

Arm	Solved/640	Exhausted	Nodes (median)	Nodes (mean)	Path len (median)	$\subseteq$ baseline?
baseline	634	6	11	1662.0	8	—
r1	619	21	13	2802.6	12	yes
r2	602	38	13	3637.1	11	yes
x	540	100	10	14680.7	9	yes
y	523	117	10	21832.7	9	yes

$\subseteq$ baseline? tests whether the arm’s solved-idx set is a subset of the baseline’s solved-idx set, computed directly from the 640-row calibration streams (not assumed).

Nodes/Path-len statistics are computed over SOLVED presentations only (budget-exhausted rows report nodes_explored == budget_nodes, which would otherwise dominate the mean).

4.3 A literature-grounded word bank fails on AK(3)

We throw the full 97-word bank (Table 1) at both AK(3) forms — the textbook presentation and the dataset-representative form — at 10^5 and 10^6 nodes: 388 runs ($97 \times 2 \times 2$), 0 solved (Figure 6, appendix). At 10^6 nodes the minimum-total-length distribution concentrates hard on the floor: the representative form gives $\{13 : 88, 14 : 2, 15 : 7\}$ and the textbook form $\{13 : 86, 14 : 2, 15 : 9\}$, against a trivial target of total length 3. Even the provably-isolatable theory words — the w_k isolation family of Theorems 6 and 7 Shehper et al. [2025] — burn the full 10^6 -node budget and stall on the floor. The AK(3) floor is 13 with a small 14–15 tail, and no family, theory-grounded or brute-force, escapes it within the searched budgets.

4.4 On solvable targets, only relator-derived words solve — and only tie the control

AK(3) returns a 0-everywhere signal, which alone cannot separate “no word helps” from “no method could help.” We therefore repeat the identical protocol on two MS presentations the 2-generator baseline *does* solve, but only after tens of thousands of nodes (Figure 7, appendix). On index 625, 4 of 98 words solve: the two controls $z = r_1$ (77,395 nodes) and $z = r_2$ (80,111), plus two `relhalf` words derived from r_1 and r_2 that match those node counts *exactly* (77,395 and 80,111) — the same searches in disguise. On index 610, 2 of 99 solve: $z = r_1$ and its `relhalf` twin, both at 61,082 nodes. Across the ~ 180 generic-family (word $\times$ target) runs, 0 solve, and the unsolved runs floor at total length 16–25. No literature-grounded word beats — or even differs from — the target’s own relators; greedy substitution does not exploit a named change of variables.

Table 4: Five attack lanes layering stable-AC structure on the base search: 0 of 16,870 Lane-D solve attempts (over 6,058 distinct candidates from 180,645 harvested quotients) and 0 of the 14 grid probes (Lanes A–C) succeed, all flooring within their searched budgets.

Lane	Method	Trials/Probes	Solved	Floor reached
A: MITM	Meet-in-the-middle ball search from AK(3)/P25 (targets = 1,177 certified stably-trivial states)	2 @ 2,000,000 nodes/side	0/2	13
B: Stable-Solver	Best-first search over substitution + stabilize + eliminate moves	6 @ 300k-800k nodes	0/6	13
C: trivial-z	n-generator (n=3,4,5) trivial-z stabilization + plain greedy substitution, rep & textbook forms	6 @ 800k- 2,000,000 nodes	0/6	14-16 (rep_n3=14, rep_n4=15, rep_n5=16, textbook_n3=14, textbook_n4=15, textbook_n5=16)
D: plateau-elim	Harvest visited stabilized states, Lemma-11 eliminate, dedupe by signed-relabel symmetry, greedy re-solve every fresh 2-gen quotient	16,870 trials / 6,058 distinct (180,645 harvested)	0/16,870	13:16,844; 19:23; 20:3
E: beam	RL Beam search with a pretrained generator PPO policy (zero-shot), widths 512 and 2048	155 + 30	0/155 + 0/30	n/a (beam search records no min-total-length floor)

Trials/Solved/floor counts for lanes A-D are computed directly from archive/campaign_grid_probes.jsonl.gz (A, B, C) and archive/campaign_trials.jsonl.gz (D), cross-checked against collect_summary.json’s authoritative totals (trials=16870, candidates=180645, grid_probes=14); all matched.

Lane D’s source-facet breakdown (trial counts): laneD:D1=5866, laneD:D2=5350, laneD:D3=5497, resolve=157.

Lane E counts are row counts of runs/beam_laneD_floor.csv (155) and runs/beam_laneD_floor_w2048.csv (30); both have solved=False on every row.

4.5 Five lanes, zero solves out of 16,870 attempts

Five attack lanes layer stable-AC structure onto the base search (Table 4, Figure 3). The consolidated campaign archive harvests 180,645 distinct Lemma-11 quotients — each stably AC-equivalent to AK(3) by construction of the harvest pipeline, verified end-to-end on the exported certificates, and none previously searched, since they exist only after a stabilize–substitute–eliminate chain — and runs 16,870 solve attempts over 6,058 distinct candidates: 0 of 16,870 attempts solved. The campaign floor histogram is $\{13 : 16,844, 19 : 23, 20 : 3\}$. Lane A (meet-in-the-middle: 2 probes at 2×10^6 nodes/side against 1,177 symmetry-expanded targets) records 0 hits and was deprioritized once all 151 in-cap catalog leaves proved plain-AC-trivial. Lane B (6 stable-move-search probes, 3×10^5 – 8×10^5 nodes) solves none; its positive controls confirm the machinery is sound — known-easy instances trivialize *through* the stabilize/eliminate moves (a corrected-family instance in 3 steps, two MS instances in 1; Table 6) — so the 0/6 on the hard probes is a search miss, not a plumbing failure. Lane C (trivial- z stabilization) floors at total length 14, 15, 16 for $n = 3, 4, 5$ — exactly one added unit of floor per contentless generator. Lane E is §4.6. A fully local reproduction — 1,937 solve attempts over a 1,006-candidate census — bottoms out at exactly 13 on every one. One coincidence deserves a line: the trivial target is total length 3 for a 3-generator state and 2 for a 2-generator state, yet both the 3-generator stabilized searches and the 2-generator quotient searches floor at the same total length 13.

4.6 Zero-shot policy transfer fails on floor states

We probe whether a learned policy escapes where greedy stalls (Figure 8, appendix). A pretrained 2-generator PPO policy with beam search from prior work Fagan et al. [2026], run zero-shot at beam width 512, solves 0 of 155 floor states, while the *same* system solves 155/155 of its own training-distribution items (mean path length 6.1, max 16). Escalating to beam width 2048 with temperature annealing on the hardest 30-state core still solves 0. This is strictly an out-of-distribution transfer probe of a fixed policy, not a statement about reinforcement learning in principle: the gap between 155/155 in-distribution and 0/155 on the floor is an OOD gap, not an RL ceiling.

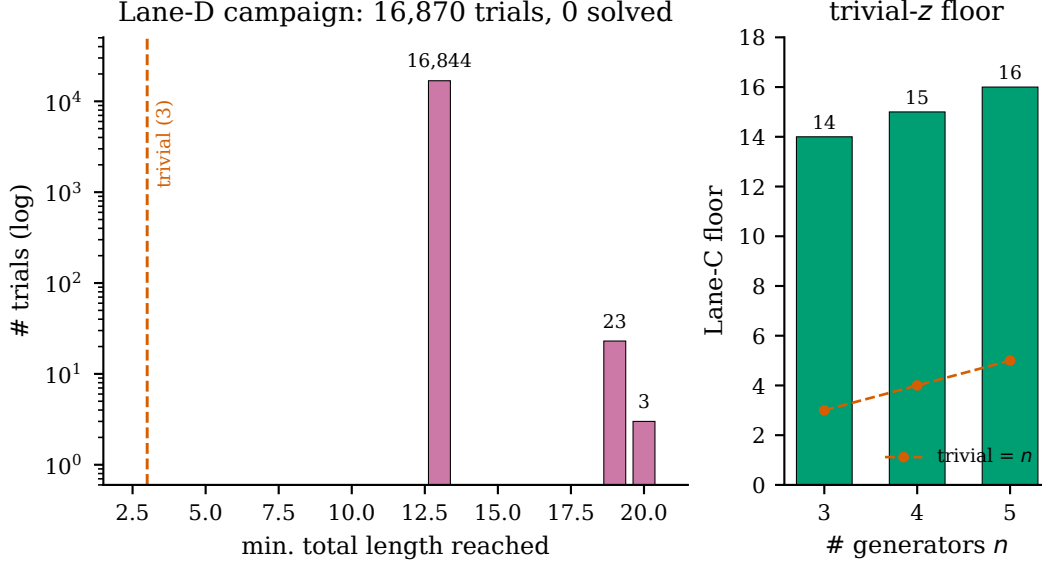


Figure 3: Floor histogram across the consolidated five-lane campaign: 16,870 solve attempts over 6,058 distinct candidates (from 180,645 harvested Lemma-11 quotients), 0 solved, with floors $\{13: 16,844, 19 : 23, 20 : 3\}$. The trivial- z control (Lane C) adds one unit of floor per contentless generator (14/15/16 for $n = 3/4/5$).

359 4.7 The cap is not the binding constraint

360 If the per-relator cap $L = 24$ were the wall, relaxing it should convert failures into solves; it does
 361 not (Figure 4, Appendix F). Raising L from 24 to 48 leaves the search’s visited set *byte-identical* in
 362 16 of 16 (form $\times$ hero-word) runs at 10^5 nodes — the search never wants a longer relator at these
 363 budgets. Harvesting with the caps raised ($l_{\text{cap}} = 48$, total-length cap 40) produces 250,397 unique
 364 candidates, of which 212,913 lie in the previously unreachable > 24 region; a stratified sample of
 365 1,800 (100 per length bucket 25–40 plus 200 short controls) solved at maximum length 60 yields 0
 366 of 1,800 attempts solved, with every single one flooring at exactly total length 13. Within the tested
 367 range the cap is not the binding constraint ($L \leq 48$, solve cap 60): the floor is a property of the
 368 region, not of the fence around it.

369 4.8 The floor has exactly two canonical representatives

370 A floor census over the 1,006 local candidates settles *where* the search funnels. Every candidate
 371 lands on one of exactly two canonical (signed-relabel) representatives of the length-13 floor of
 372 $\text{AK}(3)$ ’s AC-class (Figure 5, Tables 6, 5): $F = \langle x, y \mid y^{-2}xyx^{-2}, y^{-3}x^{-2}yx \rangle$, hit 712 times
 373 (70.8%), and $\text{AK}(3)$ ’s own reduced form, hit 294 times (29.2%). A 21-move substitution path
 374 $F \rightarrow \text{AK}(3)$ was extracted and certified, passing both the engine author’s verifier and the inde-
 375 pendently authored one (Table 6). The accessible region therefore funnels onto a *single* AC-class
 376 floor with two canonical representatives; F — the dominant attractor, reached $2.4\times$ more often than
 377 $\text{AK}(3)$ itself — is a 2-generator presentation AC-equivalent to $\text{AK}(3)$ that, to the authors’ knowledge,
 378 has not previously appeared as a search seed. The structural obstruction is now visible: the certified
 379 Appendix-F path peaks at total length 25, i.e. 12 above the floor, so escaping the floor requires
 380 climbing — precisely what length-ordered search structurally resists.

381 5 Limitations

- 382 1. **Every negative is budget-bounded.** All “unsolved” outcomes in 4 were reached within
 383 the searched budgets (2.5×10^4 to 10^6 nodes per attempt, beam width ≤ 2048), never
 384 an exhaustive one. Complexity results for balanced presentations Bridson [2015], Lishak

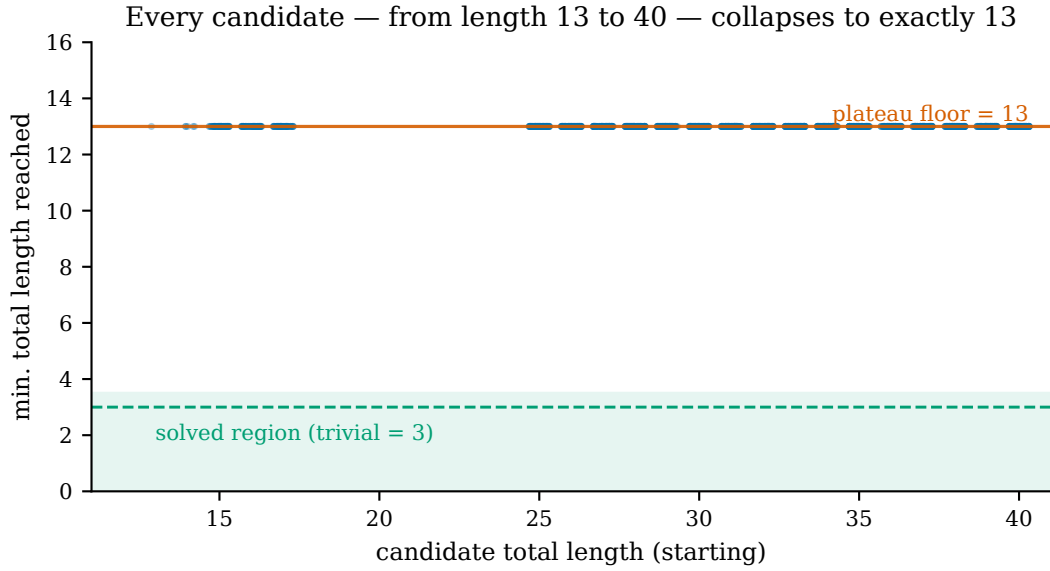


Figure 4: The per-relator cap is not the binding constraint in the tested range. Raising L from 24 to 48 leaves the visited set byte-identical in 16/16 runs at 10^5 nodes; a stratified sample of 1,800 candidates (212,913 of 250,397 raised-cap candidates lie in the previously unreachable > 24 region) solved at maximum length 60 gives 0 solved, every one flooring at exactly total length 13.

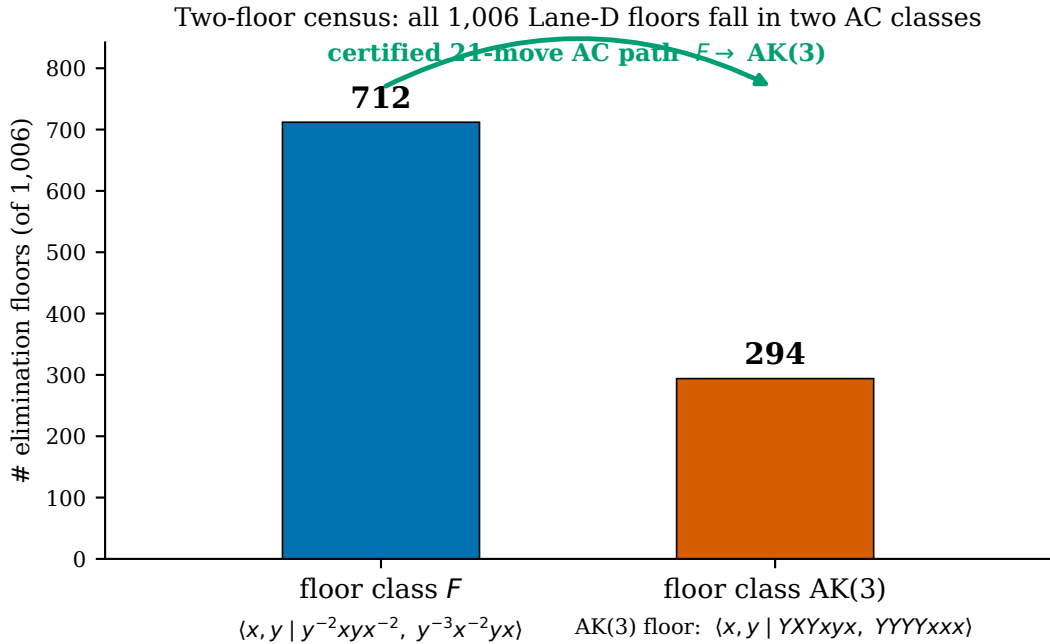


Figure 5: The length-13 floor of $AK(3)$'s AC-class has exactly two canonical (signed-relabel) representatives, over the 1,006-candidate local census: $F = \langle x, y \mid y^{-2}xyx^{-2}, y^{-3}x^{-2}yx \rangle$ (712, 70.8%; reached $2.4\times$ more often than $AK(3)$) and $AK(3)$'s own reduced form (294, 29.2%), joined by a certified 21-move substitution path $F \rightarrow AK(3)$ that peaks at total length 25, i.e. 12 above the floor.

Table 5: The length-13 floor of AK(3)’s AC-class has exactly two canonical (signed-relabel) representatives over the 1,006-candidate census: F (the dominant attractor, Class 1) and AK(3)’s own reduced form (Class 2).

Floor class	Count	Share	Relators
Class 1 (= laneF’s F)	712	70.8%	YYxyXX, YYYXXyx
Class 2 (= AK(3))	294	29.2%	YXYxyx, YYYxxxx

Floor class = the greedy search’s canonical terminal state (floor_mkey, min over signed relabelings) after re-solving every merged Lane-D quotient at a 25,000-node budget; all 1,006 probes bottom out at total relator length 13 in exactly 2 distinct classes.

Relator words translate the int-array floor_state ($1=x, -1=X, 2=y, -2=Y$).

Identity check (mitm.symmetry_keys, live): the majority class’s floor state is the laneF_F_to_AK3 cert’s F start state (mod signed relabeling); the minority class’s floor state is AK(3) itself (mod signed relabeling).

[2017] show a trivialization path, if one exists, can be far longer than any budget tested here. Read every negative as “not found within budget,” not an impossibility result.

2. **All lanes but one rank candidates by total relator length.** Lanes A–D (4) are best-first searches ordered by total relator length (or length plus a generator penalty); a globally-guided or non-monotone search was not attempted. The one full trivialization path available from prior work — the literature’s 53-move $P25 \rightarrow AK(3)$ replay — itself peaks at total length 25, above the length-13 floor these searches funnel onto, before descending to the trivial presentation. Productive paths climb before they shrink; length-ordered greedy cannot follow one by construction.

3. **The RL lane is a zero-shot out-of-distribution transfer probe, not an in-principle RL negative.** The policy was trained on 2-generator Miller–Schupp presentations and evaluated, unmodified, on 3-generator stabilized states and 2-generator floor quotients it never encountered during training. A retrained or fine-tuned policy on this distribution is future work; no conclusion about reinforcement learning’s capability in principle follows from a single zero-shot miss.

4. **Scale and provenance are stated at two levels that should not be added together.** The full-campaign totals (180,645 harvested quotients, 16,870 solve attempts, 6,058 distinct candidates) come from a consolidated, deduplicated archive verified locally; the per-box raw streams that produced it reside on cloud storage. A smaller, self-contained local reproduction (a 1,006-candidate floor census, 1,937 solve attempts) is reported separately; the two scales are kept distinct throughout.

5. **Certificate coverage is partial by design.** Five chain certificates are exported and machine-verified for the headline claims — including the 53-move $P25 \rightarrow AK(3)$ literature replay and the 21-move $F \rightarrow AK(3)$ bridge — plus 14 of 151 catalog-leaf classifications used as calibration checks; the remaining 137 leaf classifications are recorded as data but their individual certificates were not exported. The full pool of 180,645 harvested quotients is stably-AC-equivalent to AK(3) by construction of the harvest pipeline, itself verified end to end on the exported certificates, rather than individually certified one by one.

6. **The length-13 floor is not perfectly uniform.** It is the dominant outcome, but 9–23% of word-bank runs (depending on family and starting form) floor at length 14–15 instead, and 26 of the 16,870 campaign attempts floor at 19–20. “The floor is 13” summarizes a distribution, not a universal constant.

7. **The word bank is grounded in AK(3)/AK(n)’s own theory, not a target-independent theory of stabilization.** Several of its 97 words are motivated by results proved specifically for the AK(n) family Shehper et al. [2025]; applying the same bank unmodified to an arbitrary presentation is unmotivated, which is why the hard-target controls used words derived from the target’s own relators instead. A per-presentation word-bank generator is left to future work.

6 Conclusion and Next Steps

AK(3)’s stable AC-triviality remains open: the premise that had appeared to settle it was independently re-verified as rescinded in prior work Shehper et al. [2025]. Within the searched budgets ($\leq 10^6$ nodes per attempt, beam width ≤ 2048), stabilizing AK(3) in every tested form adds nothing over plain 2-generator greedy substitution — naive $z = w$ arms solve strict subsets of what the unstabilized baseline already solves, no word in the 97-word bank beats a target’s own relators as a stabilizer, and across five differently-shaped search lanes, 0 of 16,870 attempts solved. The relator-length cap does not bind in the tested range ($L \leq 48$, solve cap 60): raising it changes nothing. And the entire accessible region funnels onto a length-13 floor with exactly two canonical (signed-relabel) representatives of that floor — F , a 2-generator presentation AC-equivalent to AK(3) (71% of the census), and AK(3)’s own reduced form (29%) — joined by a certified 21-move AC path (5). That two-floor result, not a solve, is this paper’s certified structural contribution: a new fact about AK(3)’s neighborhood, established independently of whether AK(3) itself ever trivializes.

The negative and the floor structure point at the same mechanism. Greedy substitution, ordered by shrinking total length, cannot assemble a Lemma-11 change-of-variables one length-decreasing move at a time, because the cancellation the lemma provides is inherently a batch effect, not a gradual descent — the useful move is an atomic destabilization supermove, not incremental substitution. The literature’s own 53-move trivialization path confirms this directly: it peaks at total length 25 before falling to trivial, climbing before it descends. Read this way, the two-hump landscape reported for the unstable problem Fagan et al. [2026] reappears here one level up — a second hump, for the stable problem, separating the length-13 floor from the trivial presentation, that only a search capable of climbing (not just shrinking) has any chance of crossing.

Next steps.

1. Make Lemma-11 elimination itself a first-class, length-increasing search move, gated against the certified 21-move chain; the StableSolver machinery already exists (3) and needs scaling, not redesign.
2. Attack F directly. It is the dominant floor representative (71% of the census) and, unlike AK(3), was never itself used as a search seed before this work; inverting the certified $F \rightarrow \text{AK}(3)$ bridge gives a ready-made route to search outward from it.
3. Train or fine-tune the RL policy on floor quotients and stabilized states directly, closing the zero-shot out-of-distribution gap noted in §5.
4. Escalate the archived 180,645-quotient candidate pool at 10–40 $\times$ the tested budgets; its deduplicated, resumable construction (D) makes this a scaling exercise, not a rerun from scratch.
5. Test non-length-ordered or globally-guided search; the Appendix-F path’s length-25 peak is a concrete target phenomenon that a length-ordered search cannot produce by construction.
6. Build a per-presentation word-bank generator that generalizes the target-derived-relator construction used for the hard-target controls, so word choice no longer depends on AK(n)-specific theory.
7. Export certificates for the remaining 137 catalog-leaf classifications and extend the independent verifier to cover the harvest pipeline end to end, not only the certificates already exported.

Negative results of this breadth, reproducible and certificate-grade, carve the search space for anyone who takes up this problem next. The floor census, the certificate chains, and the deduplicated candidate archive are released, on acceptance, as a foundation for exactly that.

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515 A Glossary of AC Moves and Stable-AC Theorems

516 This appendix collects, in one place and with precise statements, the move definitions and theorems
517 that 3 and 4 rely on.

518 A.1 Elementary Andrews–Curtis moves

519 The Andrews–Curtis conjecture Andrews and Curtis [1965] concerns *balanced* presentations
520 $\langle x_1, \dots, x_n \mid r_1, \dots, r_n \rangle$ (equally many generators and relators). Three elementary moves act on
521 the ordered relator tuple:

- 522 • **AC1:** replace some r_i by $r_i r_j$ for $j \neq i$ (multiply one relator by another).
- 523 • **AC2:** replace some r_i by r_i^{-1} (invert a relator).
- 524 • **AC3:** replace some r_i by $g r_i g^{-1}$, where g is a generator or its inverse (conjugate a relator
525 by a single generator).

526 Two presentations related by a finite sequence of AC1–AC3 are *AC-equivalent*; a presentation AC-
527 equivalent to $\langle x_1, \dots, x_n \mid x_1, \dots, x_n \rangle$ is *AC-trivial*. The Andrews–Curtis conjecture asserts every
528 balanced presentation is AC-trivial.

529 A.2 Stable AC moves and the stable conjecture

530 The *stable* (or *weak*) AC conjecture permits two further moves:

- 531 • **AC4:** introduce a fresh generator together with the trivial relator equal to it: $\langle x_1, \dots, x_n \mid$
532 $r_1, \dots, r_n \rangle \rightarrow \langle x_1, \dots, x_n, x_{n+1} \mid r_1, \dots, r_n, x_{n+1} \rangle$.
- 533 • **AC5:** remove a trivial relator and its corresponding generator — the inverse of AC4.

534 Presentations related by AC1–AC5 are *stably AC-equivalent*; a presentation stably AC-equivalent to
535 a trivial presentation is *stably AC-trivial*. The stable conjecture asserts every balanced presentation
536 is stably AC-trivial.

537 A.3 The $z = w$ stabilization composite

538 The stabilization move used throughout 3 packages an AC4 step with the moves realizing a definition
539 $z = w(x, y)$ into a single composite:

$$\langle x, y \mid r_1, r_2 \rangle \rightarrow \langle x, y, z \mid r_1, r_2, \text{cyc_reduce}(z \cdot w(x, y)^{-1}) \rangle,$$

540 legal as a stable-AC move for *any* word w . Undoing it (destabilization) requires the target generator
541 to occur exactly once in some relator, which is exactly the precondition Lemma 11 below formalizes.

542 A.4 Lemma 11 (Substitution and Removal) — Shehper et al. [2025]

543 Let $P = \langle x_1, \dots, x_n, y \mid r_1, \dots, r_n, y^{-1}w \rangle$ present the trivial group, where w is a word in
544 $x_1, \dots, x_n$ only. Then $P' = \langle x_1, \dots, x_n \mid r'_1, \dots, r'_n \rangle$, where each r'_i is r_i with every occurrence of
545 y replaced by w , is stably AC-equivalent to P .

546 *Proof idea.* Substituting w for y throughout gives $\tilde{P} = \langle x_1, \dots, x_n, y \mid r'_1, \dots, r'_n, y^{-1}w \rangle$; since
547 $r'_1, \dots, r'_n$ contain no y , the map $x_i \mapsto x_i, y \mapsto w$ is a surjective homomorphism $\tilde{P} \rightarrow P'$, and
548 since $\tilde{P}$ is trivial, so is P' . Triviality of P' forces w to be a product of conjugates of the r'_i ; that
549 expression is precisely an AC1–AC3 sequence turning $y^{-1}w$ into y , after which y is deleted by AC5.
550 $\square$

551 The proof gives no bound on the number of conjugate factors — equivalently, no bound on the
552 number of elementary AC moves this supermove packages. Finding such a bound, or an alternative
553 proof that supplies one, remains an open question noted by Shehper et al. [2025]; every use of
554 Lemma 11 in this paper’s search is therefore a genuinely unbounded-cost supermove, not a bounded
555 shortcut.

556 A.5 The Miller–Schupp and Akbulut–Kirby families

$$\text{MS}(n, w) = \langle x, y \mid x^{-1}y^n x = y^{n+1}, x = w \rangle, \quad \text{AK}(n) = \langle x, y \mid x^n = y^{n+1}, xyx = yxy \rangle.$$

- 557 • **Theorem 2** Shehper et al. [2025]: $\text{MS}(1, w)$ is AC-trivial for every w .
- 558 • **Theorem 3**: $\text{MS}(n, w_*)$ is AC-trivial for every n , where $w_* = y^{-1}xyx^{-1}$; the proof
559 rewrites the defining relator and applies the automorphism $x \leftrightarrow y$ to reduce it to Theorem
560 2.
- 561 • **Proposition 5** (MMS02, restated in Shehper et al. [2025]): for all $n \geq 2$, $\text{AK}(n)$ is AC-
562 equivalent to $\text{MS}(n, w_1)$ with $w_1 = y^{-1}x^{-1}yxy$.
- 563 • **Theorems 6 and 7** Shehper et al. [2025]: writing $w_k = y^{-k}x^{-1}yxy$ for $k \in \mathbb{Z}$, Theorem 6
564 shows $\text{MS}(n, w_k)$ is AC-equivalent to $\text{MS}(n, w_{k+1})$ for each fixed n , so the entire family
565 $\{\text{MS}(n, w_k)\}_{k \in \mathbb{Z}}$ is one AC-equivalence class; Theorem 7 identifies each member with
566 $P(n, k) = \langle x, y \mid y^{n-k-1}x^{-1}yx = xyx^{-1}y^{n-k}, x = w_k \rangle$, of total length $|k| + |n - k| +$
567 $|n - k - 1| + 11$, minimized at $k = n - 1$. These two results are the theory grounding the
568 word bank’s **wk** family (1).

569 A.6 The MMS02 construction (Theorem 1.4)

Any presentation $\langle x, y, z \mid x = z \cdot [[y^{-1}, x^{-1}], z], y = x \cdot [[y^{-1}, x^{-1}], z^{-1}] \cdot [z^{-1}, x], w \rangle$,
570 where w is a word in x, y, z of exponent sum ± 1 , was claimed by Myasnikov et al. [2002] (MMS02)
571 to always present the trivial group. This is the theorem whose supporting construction contains the
572 misprint discussed in E.

573 A.7 Canonical form

Throughout the search and the certificates, a relator is canonicalized as the letter-order-minimal
574 representative, over all cyclic rotations of the word *and* of its formal inverse, under the total order
575 $Z < z < Y < y < X < x$ (an inverse letter precedes its generator; higher-magnitude generator
576 id first) — encoding $x \rightarrow 1, X \rightarrow -1, y \rightarrow 2, Y \rightarrow -2, z \rightarrow 3$. A presentation’s canonical key
577 sorts its canonicalized relators. Two states that agree up to this canonicalization *and* up to a *signed*
578 relabeling — a bijective, sign-respecting renaming of the generator set — are treated as equal for
579 deduplication and for cross-presentation floor comparisons (5).
580

581 B The acx-cert-v1 Certificate Schema and Its Verification

582 Every solution and every stable-AC equivalence claimed in this paper is exported as an acx-cert-v1
583 JSON certificate and checked by two independently written verifiers before it is reported.

584 B.1 Schema

585 A certificate is a JSON object with fields: `certificate_version` (schema version string); `name`;
586 `claim` (a human-readable statement of what is being certified); `start` and `end` (each an object
587 `{n_gen, relators}`); `end_is_trivial` (boolean, true iff end must equal the trivial presentation
588 on `n_gen` generators); `steps` (an ordered list of length T); `states` (the list of intermediate pre-
589 sentations, length $T + 1$, with `states[0]==start` and `states[-1]==end`); and `meta` (free-form
590 provenance — source paper or line numbers, generator naming, and similar bookkeeping).

591 B.2 Step types

592 Six step types are implemented; the field names below are exact, not paraphrased:

- 593 • **concat** `{i, j, sign}` — $r_i \rightarrow \text{freely-reduced}(r_i \cdot r_j^{\text{sign}})$ (AC1; `sign=-1` concatenates
594 with r_j^{-1}).
- 595 • **conjugate** `{i, g}` — $r_i \rightarrow \text{freely-reduced}(g \cdot r_i \cdot g^{-1})$ (AC3).

Table 6: The five exported acx-cert-v1 chain certificates, each passing both the engine author’s verifier and the independently authored black-box verifier.

Certificate	Claim	Steps	States
appendxF_P25_to_AK3	P25 is AC-equivalent to AK(3) via the 53 printed Appendix-F h-moves (arXiv:2408.15332v2).	53	54
laneB_M3corr_hero8_500	Stable-AC trivialization of M3corr_hero8 found by Stable-Solver.	3	4
laneB_ms0	Stable-AC trivialization of ms0_plain found by StableSolver.	1	2
laneB_ms0_stab	Stable-AC trivialization of ms0_stab found by StableSolver.	1	2
laneF_F_to_AK3	$F = \langle x, y \mid YYxyXX, YYYXXyx \rangle$ is plain-AC-equivalent to AK(3) (up to signed relabeling); explicit substitution path found by greedy search.	21	22

All 5 certificates pass both experiments/stable_ac/ak3_stable_proof/verify_certificate.py and the independently-authored independent_verifier.py (confirmed live: verify_certificate.py=True, independent_verifier.py=True).

- **stabilize** $\{z, w\}$ — appends generator $z = n_gen+1$ and the cyclically-reduced relator $z \cdot w^{-1}$ (AC4 plus realizing $z = w$).
- **eliminate** $\{gen, ri, \dots\}$ — Lemma-11 removal of gen using its exactly-once occurrence in relator ri ; generators above gen are renumbered down by one (the packaged AC5).
- **relabel** $\{perm, invert\}$ — a bijective, sign-respecting permutation of the generator set: an AC-equivariant renaming, not a move on relators.
- **substitution** $\{ci, a, b, i, j, c_inv\}$ — the composite move used by the base greedy solver (3); verified by checking that the resulting state lies in the recomputed neighbor set of the preceding state, rather than by replaying finer sub-steps.

There is no separate step type for a bare AC2 inversion of a whole relator. A full-relator sign flip is instead absorbed by the canonical-equality quotient the verifiers already apply (reorder relators, cyclically rotate, invert) — see the mutation-sensitivity note below, where this distinction mattered in practice.

B.3 Verification levels

- **L1 (replay).** Recompute each step’s post-state from its recorded parameters and the preceding state; it must equal the next recorded state exactly.
- **L2 (preconditions).** `eliminate`’s generator must occur exactly once in the cited relator; `relabel`’s permutation must be bijective and sign-consistent; every letter must lie in the valid signed-generator range for that state’s `n_gen`.
- **L3 (global invariants).** Every state must be balanced (`n_gen` equal to the relator count, no empty relator); the abelianization matrix’s determinant must have absolute value 1 at every state, computed by exact-integer Bareiss elimination (no floating point); if `end_is_trivial` is claimed, the end state must equal $\langle x_1, \dots, x_k \mid x_1, \dots, x_k \rangle$ up to relator reordering.

B.4 Two independent verifiers

Every certificate is checked by the engine author’s verifier and by a second verifier authored black-box from the schema and the math above alone, with no access to the engine’s source and its own free/cyclic reduction, canonicalization, integer-determinant, and step-replay logic. Across the campaign the two verifiers together performed 20,947 checks with 0 failures. All five exported chain certificates pass both (Table 6): `appendxF_P25_to_AK3` (53 steps, the literature’s $P25 \rightarrow AK(3)$ replay of E), `laneF_F_to_AK3` (21 steps, the path from F — a 2-generator presentation AC-equivalent to $AK(3)$ — to $AK(3)$ itself), and three small stable-solver certificates recording in-search destabilizations (3, 1, and 1 steps respectively).

B.5 Mutation sensitivity

Sign flips, corrupted target states, swapped intermediate states, and mis-pointed eliminations are all rejected by both verifiers. One candidate mutation was initially flagged as accepted; on inspection it was not a soundness hole but a genuinely valid alternative move — inverting a relator of length one is itself a legal AC2 move, correctly treated as a no-op by the canonical-equality quotient rather than as a tamper. A semantically meaningful tamper test therefore has to flip an *interior* letter of a relator of length at least three in a middle state, not merely a sign.

B.6 Coverage and chain composition

Of the 151 catalog leaves classified during the campaign, 14 have exported, machine-verified leaf certificates; the remaining 137 classifications are recorded as data (solved/unsolved, node count, floor reached) but were not individually exported as certificates. Certificates concatenate whenever one certificate’s end state matches the next certificate’s start state exactly, so a full stable-AC chain — stabilize → substitution path → eliminate → 2-generator trivialization path, further composed with the P25 bridge of E — assembles automatically into a single machine-checkable certificate. This composition machinery already produces every certificate reported above and would apply unchanged to any future solve.

C Extended Data Tables

This appendix collects extended data behind the six numbered tables of the main text (Tables 2–5) and discusses the three figures that live only in the appendix (6, 7, 8). Every value below is read from the same digest files that produced the main-text tables and figures, and is independently re-derivable from the raw record streams (D).

C.1 Extended per-arm statistics

Table 3 reports median nodes and median path length, over solved presentations only, for the naive $z = w$ stabilization arms. The full min/max/mean/median breakdown, over the same solved-only convention, is:

Arm	Nodes (min / max / mean / median)	Path length (min / max / mean / median)
baseline	3 / 77,385 / 1,662.0 / 11	2 / 710 / 32.6 / 8
r_1	5 / 239,578 / 2,802.6 / 13	4 / 714 / 33.8 / 12
r_2	4 / 291,479 / 3,637.1 / 13	3 / 715 / 29.9 / 11
x	4 / 326,326 / 14,680.7 / 10	3 / 102 / 12.1 / 9
y	4 / 491,437 / 21,832.7 / 10	3 / 102 / 11.5 / 9

The x and y arms’ wide gap between median (10) and mean (14,680.7 / 21,832.7) node counts, set against a comparatively tight path-length distribution (max 102), is the alias-orbit effect noted in 4: defining z by an already-present generator does not lengthen solutions, it multiplies equivalent detours the search must exhaust before terminating.

C.2 The 97-word literature-grounded bank

Table 1 gives family sizes. The full bank — 95 words across seven families, plus the two per-target control words $z = r_1, r_2$ added at run time to reach 97 — grouped by family in escalation-priority order, is:

- **relhalf** (17): $xyx, yxy, xxx, yyyy, yxx, xxy, xyy, yyx, YXX, YXY, XXX, YYYY, xyxY, Yxyx, xx, yy, yyy$
- **wk** (17): $yyyyyyyyXyxy, yyyyyyXyxy, yyyyyyXyxy, yyyyyXyxy, yyyyyXyxy, yyyXyxy, yyXyxy, yXyxy, Xyxy, YXyxy, YYXyxy, YYYXyxy, YYYYYXyxy, YYYYYXyxy, YYYYYXyxy, YYYYYXyxy$
- **wstar** (5): $YxyX, XyxY, xYXy, yXYx, xYxY$

- **conj** (14): $x, xyX, yxY, y, Xyx, Yxy, xyxYX, yxyXY, xYxyX, YxyXy, xxyXX, yyxYY, XYxyx, XyxYx$
- **comm** (6): $XYxy, YXyx, xyXY, yxYX, XyXy, xyXYX$
- **ms** (3): $Xyxy, yxyx, xyxy$
- **brute** (33): $X, Y, xy, xY, XX, Xy, XY, yx, yX, Yx, YX, YY, xxY, xYx, xYX, xYY, XXy, XXY, XyX, Xyy, XYx, XYY, yXX, yXy, yXY, yyX, Yxx, YxY, YXX, YXy, YYx, YYX, YYY$

Each name is the induced word w in $x^{\pm 1}, y^{\pm 1}$ under the encoding of A (capital letter = inverse); each induces a z -relator $z \cdot w^{-1}$, deduplicated by that induced relator and dropped if it violates the per-relator cap $L = 24$.

C.3 Campaign facet breakdown

Lane D's 16,870 solve attempts (Table 4) are drawn from four resumable facets, every one contributing 0 solved: D1 (5,866 solve attempts), D2 (5,350), D3 (5,497), and a resolve pass at the raised per-relator cap $L = 40$ (157 solve attempts, probing whether the $L = 24$ cap specifically — as opposed to the total-length cap — was pruning a viable descent among the shortest candidates). The four facets sum to the reported 16,870; 0 of 16,870 solve attempts succeed across all four.

C.4 Floor census detail

The two canonical (signed-relabel) representatives of the length-13 floor of $AK(3)$'s AC-class (Table 5) have relators, under the encoding $x \rightarrow 1, X \rightarrow -1, y \rightarrow 2, Y \rightarrow -2$:

- F — a 2-generator presentation AC-equivalent to $AK(3)$: $YYxyXX / YYYXXyx$, i.e. $y^{-2}xyx^{-2}$ and $y^{-3}x^{-2}yx$ — the dominant attractor, 712/1006 (70.8%).
- $AK(3)$'s own reduced form: $YXYxyx / YYYYxxx$ — 294/1006 (29.2%).

Each census candidate is assigned by canonicalizing its terminal state and checking equality with one of these two representatives up to signed relabeling (A); no third representative appears anywhere in the 1,006-candidate census.

C.5 Grid-probe list

Beyond Lane D's harvested-quotient attempts, the campaign ran 14 additional grid probes (Table 4, Lanes A–C):

Lane	Probe	Nodes	Floor
A (MITM)	from $AK(3)$, against 1,177 targets	2,000,000	13
A (MITM)	from P25, against 1,177 targets	2,000,000	13
B (StableSolver)	hero-8 bank, $g \leq 3$, gen-penalty 1	300,000	13
B (StableSolver)	hero-8 bank, $g \leq 3$, gen-penalty 2	300,000	13
B (StableSolver)	hero-8 bank, $g \leq 4$, gen-penalty 1	300,000	13
B (StableSolver)	hero-8 bank, $g \leq 4$, gen-penalty 2	300,000	13
B (StableSolver)	full 95-word bank, from $AK(3)$	100,000	13
B (StableSolver)	full 95-word bank, from P25	100,000	13
C (trivial- z)	rep form, $n = 3$	800,000–2,000,000	14
C (trivial- z)	rep form, $n = 4$	800,000–2,000,000	15
C (trivial- z)	rep form, $n = 5$	800,000–2,000,000	16
C (trivial- z)	textbook form, $n = 3$	800,000–2,000,000	14
C (trivial- z)	textbook form, $n = 4$	800,000–2,000,000	15
C (trivial- z)	textbook form, $n = 5$	800,000–2,000,000	16

All 14 probes floor within their searched budgets; 0 of 14 solve.

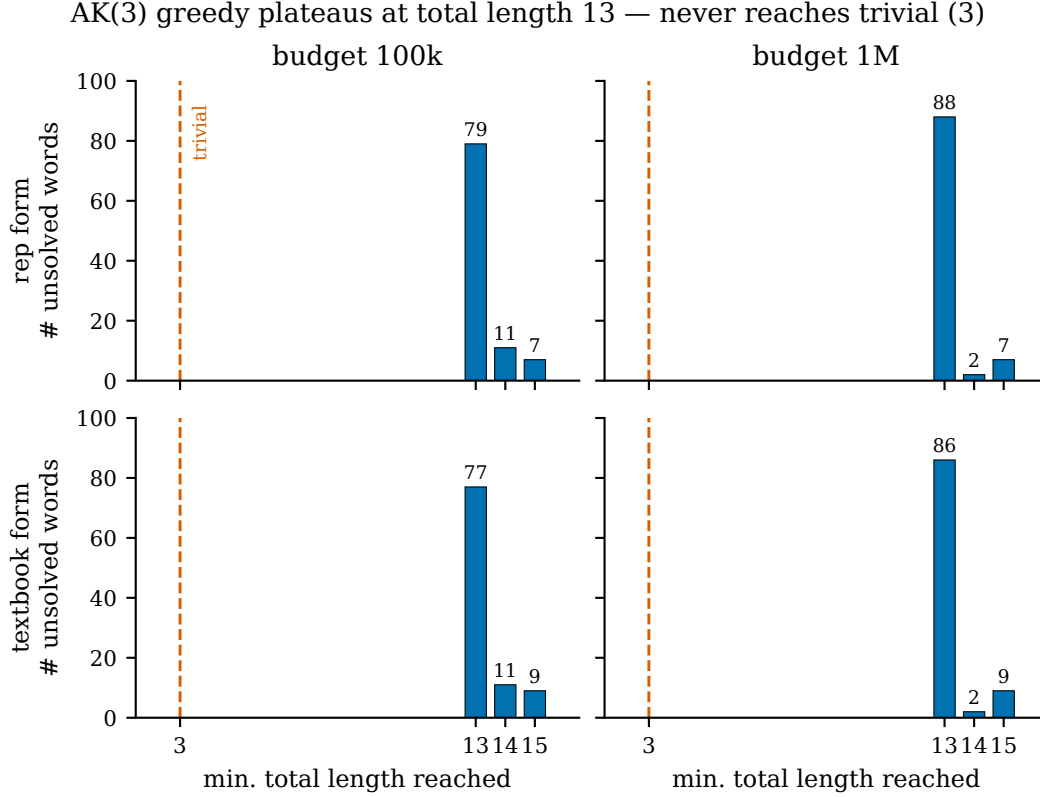


Figure 6: Minimum-total-length reached by the 97-word bank on both AK(3) forms (trivial target = 3); 0 of 388 solve attempts succeed. At 10^6 nodes the distribution concentrates on the floor — representative form $\{13:88, 14:2, 15:7\}$, textbook form $\{13:86, 14:2, 15:9\}$ — a floor of 13 with a small 14–15 tail that even the provably-isolatable w_k theory words do not escape.

700 C.6 Discussion of the appendix figures

701 **Figure 6.** At 10^5 nodes the representative form’s floor histogram is already concentrated (79 words
702 at floor 13, 11 at 14, 7 at 15); escalating every unsolved word tenfold to 10^6 nodes tightens the
703 concentration further (88/2/7) rather than dislodging any word past the floor, and the textbook
704 form shows the identical pattern (77/11/9 at 10^5 , sharpening to 86/2/9 at 10^6). Every one of the
705 $97 \times 2 \times 2 = 388$ solve attempts across both forms and both budgets records 0 solved, within the
706 searched budgets, against a trivial target of total length 3; more nodes concentrate the distribution
707 on the floor rather than escaping it, evidence that the obstruction is structural rather than a matter of
708 search depth.

709 **Figure 7.** On the two hard-but-solvable controls, the floor histograms sit far above AK(3)’s own
710 (index 625: range 17–25, peaking at 19 with 41 of 98 words; index 610: range 16–23, peaking at
711 19 with 34 of 99 words), confirming these are genuinely different, easier landscapes with their own
712 internal spread. Yet the only words that solve are, in both cases, exactly the target’s own relators
713 and their relator-derived *relhalf* twins, and they solve at identical node counts (index 625: r_1 and
714 its *relhalf* twin both at 77,395 nodes; r_2 and its twin both at 80,111; index 610: r_1 and its twin
715 both at 61,082) — the same search in disguise, not a shortcut. Within the searched 10^5 -node screen
716 (the specified 10^6 -node full tier was not run on these generic-family words), no structurally distinct
717 word ever solves either target.

718 **Figure 8.** The pretrained policy’s own training-distribution path lengths are short and tightly clustered
719 (mean 6.1, range 2–16, mode at length 4: 31 of 155 items), and it solves all 155 of 155 solve
720 attempts on that distribution. Run zero-shot on the 155 floor and stabilized states at beam width 512,
721 0 of 155 solve attempts succeed; widening to beam width 2048 with temperature annealing on the
722 hardest 30-state core, 0 of 30 solve attempts succeed. Within these two beam-search configurations,

Only relator-derived z solve, and they tie the $z = r_1$ control's node count

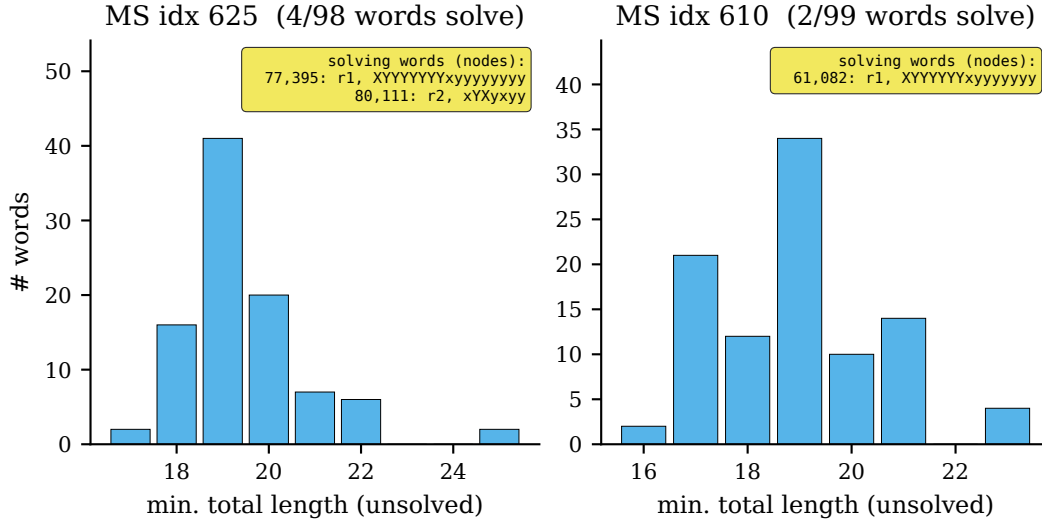


Figure 7: Word-bank protocol on two hard-but-solvable MS targets. Index 625: 4 of 98 words solve — controls $z = r_1$ (77,395 nodes) and $z = r_2$ (80,111) plus two relator-derived `relhalf` words tying those exact node counts. Index 610: 2 of 99 solve ($z = r_1$ and its `relhalf` twin, both 61,082). All ~ 180 generic-family runs fail (floor 16–25); no clever word beats or even differs from the target’s own relators.

Agent solves all training presentations but 0 of the 155-floor eval set

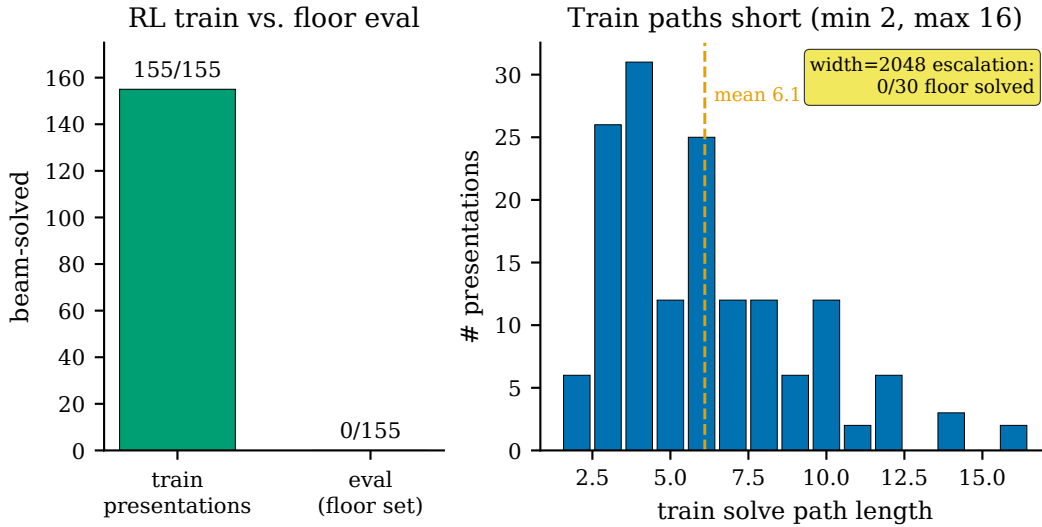


Figure 8: Zero-shot out-of-distribution transfer of a pretrained 2-generator PPO+beam policy. It solves 155/155 of its own training-distribution items (mean path 6.1, max 16) but 0 of 155 floor states at beam width 512; escalation to beam width 2048 with temperature annealing still solves 0 of the hardest 30. The gap is an OOD gap, not an RL ceiling.

the size of this gap — perfect in-distribution, zero out-of-distribution, on states sharing the same 2-generator alphabet the policy was trained on — is the sharpest illustration in the campaign that the floor resists every method family tried, learned or hand-designed, within the budgets searched.

D Reproducibility and Artifact Details

D.1 Data and artifacts

- **MS(1190) dataset.** The 640-presentation known-solvable prefix used throughout 3, drawn from the Miller–Schupp benchmark Miller and Schupp [1999].
- **Consolidated candidate/trial archives.** JSONL.gz streams with one record per (presentation, arm/word/candidate, budget) triple and full per-record provenance — source form, generator and relator words, node count, floor reached, solved flag.
- **Certificates.** JSON, schema of B.

All of the above, together with the two acx-cert-v1 verifiers, are released on acceptance. The source MS(1190)/AC-benchmark datasets this campaign builds on are released under a data license (CC BY 4.0) by Fagan et al. [2026]; the campaign’s own derived artifacts — word bank, quotient archive, certificates — will be released under the same terms.

D.2 Per-campaign recipe

Six campaigns, each an independent, resumable, append-only JSONL stream keyed by (dataset, idx-or-form, word-or-arm, budget):

1. **Baseline reproduction** (cap-semantics check, 3). Entry: `run_greedy.py`. Parameters: `-cap {sum,per_relator} -budget {1e5,1e6} -workers N`. Output: one stream per (cap, budget) tier, one record per presentation index, distinguishing the per-relator cap ($L = 24$) from the original sum cap.
2. **Naive $z = w$ arms** (Table 3). Entry: `run_greedy_sweep.py`. Parameters: `arm  $\in$  {baseline,  $r_1, r_2, x, y$ }, budget  $5 \times 10^5$` , two-phase light/heavy worker scheduling so cheap instances finish first and memory-heavy ones run at low parallelism. Output: one stream per arm, records keyed by (dataset, idx, arm, budget) so arms merge directly for the subset/union comparison of 4.
3. **AK(3) word bank** (Table 1). Entry: `run_ak3_wormhole.py`. Parameters: `form  $\in$  {textbook, rep}`, the 97-word bank, a two-tier budget (10^5 -node screen, escalate every unsolved word to 10^6), per-relator cap $L = 24$. Output: one stream per (form, budget), named by the pattern `ak3_<form>_<budget>.jsonl`, keyed by word name.
4. **Hard-target controls** (Figure 7). Entry: `run_hard_wormhole.py`. Parameters: MS(1190) indices `{625,610}`, target-specific word banks (the `relhalf` family re-derived from each target’s own relators), 10^5 -node screen. Output: one stream per target index, keyed by word name.
5. **Five-lane campaign** (Table 4). Entries: `plateau_elim.py` (Lane D’s harvest/merge/solve phases), with `harvest_fast.py` as its numba-accelerated, byte-identical hot path; `stable_solver.py` (Lane B); `floor_census.py` (the two-floor census); `resolve_hi_1.py` (cap-gap re-solve of the shortest candidates at a raised per-relator cap); `collect_results.py`, which consolidates every box’s streams into one archive plus a manifest. Parameters vary by phase: harvest budgets up to $\sim 5 \times 10^5$ per combination, solve budgets 2.5×10^4 – 2×10^5 , `resolve_hi_1.py` at cap $L = 40$. Output: per-combination candidate streams, a global merged-candidate stream, a solve-attempt stream, and certificate JSON files — keyed by (form, word) for harvest and by candidate canonical key for merge/solve.
6. **Cap control** (F). Entries: `run_capttest.py` (positive-control gate, the search-cap arm, and the harvest-cap arm) and `solve_stratified.py` (the stratified-solve arm). Parameters: gate at 5×10^4 nodes; search arm at `max_len  $\in$  {24,48}`, 10^5 nodes; harvest arm at per-relator cap 48, total-length cap 40, 6×10^4 nodes; stratified solve at 100 candidates per length bucket 25–40 plus 200 short controls, `max_len = 60`, budget 8×10^4 (4×10^4 for

773 the longest-tail bucket). Output: isolated JSONL streams, kept entirely separate from the
774 main campaign’s directories.

775 Every stream is append-only, one JSON record per line, flushed (and fsynced on cloud/networked
776 mounts) after each write. A re-run of a finished sweep re-reads existing ids and exits as a no-op; a
777 killed run resumes from the last complete line, discarding at most one in-flight record.

778 D.3 Determinism and verification

779 The per-node hot paths — neighbor generation, canonicalization, Lemma-11 elimination — are
780 numba-jitted. Every jitted path is held byte-identical to a pure-Python reference implementation
781 via differential tests on real visited sets, with 0 mismatches across the tested combinations. Every
782 claimed solve is replayed through an independent path verifier before it is counted (Table 6, B), and
783 every printed number in the paper is re-derived by an independent recomputation script run directly
784 against the raw JSONL/gz streams, not against any intermediate digest.

785 D.4 Compute

786 Development and calibration ran on a single consumer laptop (16 GB RAM); the five-lane escalation
787 additionally used cloud CPU boxes (up to 8 vCPU, up to 50 GB RAM); the RL beam probe (Lane
788 E) ran as CPU-only inference throughout. Per-attempt node budgets ranged from 2.5×10^4 to 10^6 ; a
789 greedy run’s visited set carries roughly $40 \times$ its node budget in entries, which is the binding memory
790 constraint at the higher budgets. Total wall-clock across the full campaign was approximately one
791 week.

792 E The MMS02 Misprint and Its Consequences

793 E.1 The construction, and where it breaks

794 Theorem 1.4 of MMS02 Myasnikov et al. [2002] builds a family of trivial-group presentations from
795 a Wirtinger presentation of a specific unknot diagram: a 14-generator, 14-relator presentation in
796 which, as for any Wirtinger presentation, every single relator is redundant, so deleting any one
797 relator still presents the knot group — the infinite cyclic group $\mathbb{Z}$ for the unknot. Deleting one
798 chosen relator and adjoining a further relator w of exponent sum ± 1 in the remaining generators,
799 then recursively eliminating generators via the Lemma-11 substitution-and-removal move (A), was
800 claimed to always yield a trivial-group presentation on three generators.

801 The construction’s premise fails at the source. As printed, the 13th Wirtinger relator reads $x_{13} =$
802 $x_5 x_{12} x_5^{-1}$; the diagram it is meant to transcribe requires $x_{13} = x_4 x_{12} x_4^{-1}$. This is not cosmetic: in
803 a genuine Wirtinger presentation, deleting any single relator must leave the same group, but deleting
804 different relators from the misprinted 14-relator presentation W' gives *different* groups — deleting
805 relator 7 leaves a presentation admitting 12 homomorphisms to S_3 (of which 6 are non-abelian,
806 the fingerprint of the three-strand braid group B_3), while deleting relator 14 leaves a $\mathbb{Z}$ -consistent
807 presentation. No Wirtinger presentation of any knot diagram can exhibit that disagreement, so W' is
808 not a Wirtinger presentation of any diagram, Theorem 1.4’s argument does not apply to the printed
809 family, and the stable AC-triviality of the length-25 presentation P25 derived from it — and hence
810 of AK(3), which is AC-equivalent to P25 — loses its only support. This gap was first reported by
811 Shehper et al. [2025] (v2, Appendix F), which rescinded the stable-triviality claim its own v1 had
812 made.

813 Seven independent computational checks reproduce every piece of this account (Table 2): all 14
814 single-relator deletions of the corrected Wirtinger presentation present $\mathbb{Z}$; the misprinted deletions 7
815 and 14 disagree exactly as described; AK(3) itself presents the trivial group; the Wirtinger-derived
816 reduction matches P25 under the pinned commutator convention; and the corrected family’s 3-
817 generator reduction presents $\mathbb{Z}$.

818 E.2 The surviving plain-AC bridge

819 The misprint does not touch the *plain* AC-equivalence $P25 \sim AK(3)$, witnessed by a 53-move path
820 printed in Appendix F of Shehper et al. [2025]. Replayed here under all four order/inversion con-

ventions available for the path’s constituent moves, only the forward-order, forward-move reading lands exactly on $AK(3)$ — equal as ordered word pairs, with no canonicalization needed — confirming both the transcription of the path and the underlying move semantics. The absolute value of the abelianization determinant is 1 at every one of the 53 intermediate states, and the path’s peak total relator length is 25 — twelve above the length-13 floor discussed in the main results — so the presentation must grow before it can shrink to trivial.

827 E.3 An independently published bridge, with two data defects

828 A second, independently derived AC path between the same endpoints was published as a 159-
829 transition automated-deduction sequence, “S2” Lisitsa [2025, 2024b]. Replayed at cyclic-word
830 granularity, every transition validates as a legal AC move once two data defects in the published
831 artifact are handled: one transition is corrupted (bridged by composing the two adjacent moves,
832 which checks out); one conjugation step’s printed label gives the wrong direction (the true conju-
833 gator is the other side of the printed pair). With those two defects bridged, S2 is a genuine second
834 witness to $P25 \sim AK(3)$. Its authoring paper Lisitsa [2025], however, was written between the
835 v1 and v2 revisions of Shehper et al. [2025] and justifies P25’s stable triviality by citing the same,
836 now-rescinded Theorem 1.4; it has not since been revised or corrected. Its plain-AC contribution —
837 an independent trivialization sequence bridging P25 to $AK(3)$ — stands regardless; only the stable-
838 triviality conclusion it inherits from MMS02 is affected.

839 F Cap-Control Experiment

840 Four arms test whether the per-relator cap $L = 24$ used throughout the campaign is itself the reason
841 stabilized search stalls, run in an isolated harness that never edits the shared solver’s default cap; the
842 lifted cap is passed only as a search-time parameter.

843 **Arm 1 (positive-control gate).** Before any negative here is trusted, a known-trivial MS(1190)
844 instance (index 0) is stabilized with $z = x$ and re-solved by the 3-generator greedy search at
845 the default cap. The run must both solve and pass path verification. It does, confirming the
846 solve+stabilize+verify chain is live in the isolated harness before any negative result is reported.

847 **Arm 2 (search cap raised, $L = 24 \rightarrow 48$).** Sixteen (form $\times$ hero-word) configurations — both
848 $AK(3)$ forms crossed with eight literature-derived hero words — are searched at 10^5 nodes under
849 both $L = 24$ and $L = 48$; 0 of 32 runs solve. The two runs’ full visited sets are byte-identical in 16
850 of 16 configurations: the greedy-by-total-length search never once wants a relator longer than 24 at
851 this budget, so raising the search cap is a provable no-op here. Floors are 13 or 15 depending on the
852 word, never on the cap value.

853 **Arm 3 (harvest/elimination cap raised).** Re-running the Lane-D harvest and merge with the
854 elimination-side caps raised (per-relator cap 48, total-length cap 40) produces 250,397 unique can-
855 didates (mod symmetry), of which 212,913 have total length exceeding 24 — a region the $L = 24$
856 pipeline could never generate at all.

857 **Arm 4 (stratified solve of the newly reachable region).** Because Lane D’s own solve order is
858 shortest-first and never reaches the long tail, a separate stratified sample — 100 candidates from
859 each length bucket 25–40, plus 200 short controls — is solved directly at $\max_len = 60$, budget
860 8×10^4 nodes (4×10^4 for the longest-tail bucket, a reduction justified by the fact that no candidate
861 among more than 1,100 prior solve attempts had ever dipped below floor 13, so a smaller budget
862 cannot mask a solve that would first have to descend through that same floor). Of 1,800 solve
863 attempts, 0 solve, and every single one floors at exactly total length 13, uniformly across every
864 length bucket from 25 to 40 and the short controls alike.

865 **Scoped conclusion.** Within the tested range ($L \leq 48$, solve cap 60), lifting the per-relator cap
866 changes nothing: the search neither wants the extra room (Arm 2) nor benefits from the longer
867 candidates it makes reachable (Arms 3–4). The obstruction is a property of the length landscape
868 stabilized greedy search explores, not of the fence bounding how long a relator may grow; this
869 control does not — and cannot — rule out a fundamentally different, non-greedy search succeeding
870 somewhere in the newly reachable region.

G NeurIPS Paper Checklist

1. **Claims.** Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?
[Yes] The abstract and introduction’s claims — the re-verified misprint and rescission, the stabilization negatives at stated budgets, the cap-not-binding control, and the two-representative floor structure — are each backed by a dedicated results subsection and by machine-checkable certificates (B).
2. **Limitations.** Does the paper discuss the limitations of the work performed?
[Yes] A dedicated limitations section lists seven explicit items: the budget-boundedness of every negative, the length-ordered-search restriction, the RL lane’s out-of-distribution scope, the two-scale reporting convention, partial certificate coverage, floor non-uniformity, and the AK(3)-specific word bank.
3. **Theory assumptions and proofs.** For all theoretical results, does the paper state the full set of assumptions and include a complete proof?
[Yes] The paper proves no new theorems. Every stable-AC-equivalence or triviality claim is instead a machine-checked certificate against the acx-cert-v1 schema (B), independently verified by two separately authored programs, in place of a hand proof.
4. **Experimental result reproducibility.** Does the paper fully disclose all the information needed to reproduce the main experimental results?
[Yes] D gives, per campaign, the entry point, parameters, output stream, and resume semantics; every accelerated hot path is gated against a reference implementation.
5. **Open access to data and code.** Does the paper provide open access to the data and code, with sufficient instructions to reproduce the main results?
[Yes] Code, the consolidated result archives, and all certificates will be released on acceptance (withheld only during anonymized review); D documents the recipe needed to regenerate every reported number from them.
6. **Experimental setting/details.** Does the paper specify all the training/evaluation details needed to understand the results?
[Yes] Node budgets, caps, worker counts, and beam-search widths and temperatures are given per campaign in 3 and D.
7. **Experiment statistical significance.** Does the paper report error bars or other appropriate measures of statistical significance?
[NA] Every reported outcome is a deterministic search over a fixed, exactly enumerated set of presentations or candidates (solved counts, floor histograms, certificate check counts); there is no sampling variance to report error bars over.
8. **Experiments compute resources.** Does the paper disclose the compute used for the experiments?
[Yes] D states the hardware (a 16 GB laptop; cloud CPU boxes up to 8 vCPU / 50 GB), per-attempt node budgets, and total wall-clock (about one week).
9. **Code of ethics.** Does the research conform to the NeurIPS Code of Ethics?
[Yes] The work is a pure-mathematics search over group presentations; it involves no human subjects, personal data, or dual-use models.
10. **Broader impacts.** Does the paper discuss both potential positive societal impacts and negative societal impacts?
[Yes] This is a pure-mathematics search on the Andrews–Curtis conjecture, with no foreseeable negative societal impact identified. The practical benefit is methodological: a rigorously verified negative and a released, deduplicated candidate archive let future attempts skip already-searched regions rather than re-spend the compute this campaign already spent.
11. **Safeguards.** Does the paper describe safeguards for the responsible release of data or models with a high risk for misuse?
[NA] The released artifacts are group-presentation data, search logs, and machine-checkable certificates; none carries a misuse risk that would require a release safeguard.

- 924 12. **Licenses.** Are the creators or original owners of assets used in the paper properly credited,
925 and are the license and terms of use explicitly mentioned?
926 **[Yes]** The Miller–Schupp benchmark presentations and related dataset infrastructure are
927 drawn from prior work Fagan et al. [2026], credited at first use and released under a CC
928 BY 4.0 data license; the classical AC/AK/MS literature results are cited throughout.
- 929 13. **New assets.** Are new assets introduced in the paper well documented, and is the documen-
930 tation provided alongside the assets?
931 **[Yes]** The acx-cert-v1 certificate schema (B), the 97-word literature-grounded word bank
932 (C), and the consolidated candidate/trial archives (D) are each documented with their
933 schema, provenance, and generation recipe, and released alongside the code on acceptance.
- 934 14. **Crowdsourcing and research with human subjects.** For crowdsourcing experiments or
935 research with human subjects, does the paper include the full text of instructions and screen-
936 shots, if applicable?
937 **[NA]** This work involves no crowdsourcing and no human subjects.
- 938 15. **IRB approvals.** Does the paper describe potential risks and confirm that the research
939 received Institutional Review Board (IRB) approvals, if applicable?
940 **[NA]** No human subjects are involved in this research.